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Assessing the impact of clean cooking energy on mental health in Indonesia

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Abstract
The health consequences of household air pollution caused by the use of solid cooking fuels are a significant priority in the worldwide endeavor to promote a shift towards clean energy. In comparison to other developing countries, the transition to clean cooking energy in Indonesia is the most significant. So far, no study has attempted to quantify the causal effects of clean cooking energy adoption on mental health in Indonesia. By using Indonesian Family Life Survey (IFLS) data and complementary estimation techniques, such as Propensity Score Matching and Difference-in-Difference (PSM-DID), this study attempts to close a research gap by analyzing the effects of the cooking energy transition on mental health in Indonesia. These techniques help to overcome biases resulting from selection mechanisms that might make it difficult to identify causal effects. The findings revealed that individuals who converted to clean cooking energy had lower Center for Epidemiological Studies Depression (CESD) scores and a 2.9% lower risk of having depressed symptoms. Women and those who live in urban were more affected than other groups, with a 3.5% and 5.2% decrease in the likelihood of depressive symptoms. These results imply that switching cooking energy sources can might improve individual's mental health. This study also addresses the potential channeling of the clean cooking energy transition and mental health, using physical health status and social participation.

Keywords Clean cooking energy · Mental health · CESD · PSM-DID

JEL Classification D12 · I15 · I31 · Q40

1 Introduction

Cooking has a powerful ability to bring families closer together and has important cultural and social effects on many different countries around the world. The provision of cooking energy is imperative for fulfilling fundamental human survival requirements. Lacking access to clean cooking energy faces the potential consequence of being deprived of the opportunity to lead healthy lifestyles [1]. Nevertheless, the persistent challenge of accessing clean fuels and cooking technologies has spanned a period of thirty years, resulting in various consequences related to health, gender, the economy, the environment, and climate change. In developing countries, there remains a prevalent reliance on solid

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biomass fuels, such as firewood, coal, and kerosene, for primary cooking. According to the International Energy Agency's [2] findings, a staggering number of over 2.5 billion individuals lack access to clean cooking energy.

The incomplete combustion of conventional cooking fuels has the potential to generate household air pollution, primarily due to the presence of particulate matter PM2.5 and carbon monoxide CO [3, 4]. Household air pollution, mainly stemming from cooking smoke, has been conclusively linked to approximately 2.5 million premature deaths each year due to its adverse health effects [2]. Several studies have endeavored to establish a correlation between the utilization of unclean energy sources for cooking purposes and various health hazards, including respiratory ailments, cardiovascular disorders, pulmonary diseases, ischemic heart conditions, and compromised overall health status [5–8]. In addition to the adverse effects on physical health, the utilization of solid fuel has been demonstrated to have detrimental effects on mental health, specifically manifesting as heightened depressive symptoms [9–11].

The importance of public awareness concerning mental health has been underscored by the Global Burden of Disease study, which has elevated this topic to a prominent position in the field of global public health. The World Health Organization (WHO) identifies depression and anxiety as the prevailing mental health conditions, putting them second in terms of the most significant factors that lead to disability. Mental health disorders, particularly depression, have been reported to impact an estimated 280 million individuals worldwide. The Indonesian Ministry of Health (2017) reports that the prevalence of depression in the country has escalated to approximately 9 million individuals or 3.7% of the total population.

In light of the negative health consequences associated with the use of dirty cooking fuels, global and national efforts have been made to promote the adoption of clean cooking fuels in developing countries. Within Sustainable Development Goal (SDG) 7, the World Health Organization (WHO) has established as a primary objective the global availability of clean, affordable, dependable, sustainable, and modern cooking energy. Several countries, such as China, India, Pakistan, Ghana, and Indonesia, have successfully undertaken the clean energy transition. The Kerosene to Liquid Petroleum Gas (LPG) Conversion Program was launched by the Indonesian government in 2007. This program has successfully promoted cleaner cooking fuels and social change. According to WHO data, more Indonesians have access to clean cooking fuel. In particular, from 23% in 2007 to 63% in 2014 and 85% in 2020.

The transition to clean cooking energy has a positive impact on individual health outcomes, specifically in terms of reducing the incidence of diseases such as hypertension, lung disease, and asthma [8, 12]. The comprehensibility of this statement stems from the fact that the utilization of clean fuels has the potential to enhance the quality of indoor air by reducing the presence of detrimental pollutants that contribute to indoor air pollution. Moreover, the utilization of clean fuels in cooking has been shown to have a direct positive impact on individuals' well-being, enhancing their overall happiness, comfort, and satisfaction. This is primarily attributed to the reduction in time allocated to gathering solid fuels, such as firewood, as well as the time saved during the cooking process itself [13–15]. Hence, apart from its influence on enhancing physical health, the transition to clean cooking energy is anticipated to have a positive effect on mental health as well. Thus, it is essential to analyze the causal effects of the clean cooking energy transition on mental health in addition to its effects on physical health. According to Fig. 1, Indonesia exhibits the most substantial growth in clean cooking energy transition initiatives in comparison to other developing countries.

Previous research has examined the effect of switching to clean cooking energy on mental health. However, most of these studies have concentrated on case studies in a developed country, such as China [10, 16, 17]. Based on the available literature, the extent of research utilizing case studies in developing countries, such as Indonesia, remains significantly constrained. The majority of the available empirical research on the impacts of cooking energy relies on observational studies or cross-sectional data, which may conflate the causal effects of indoor air pollution with the determinants of

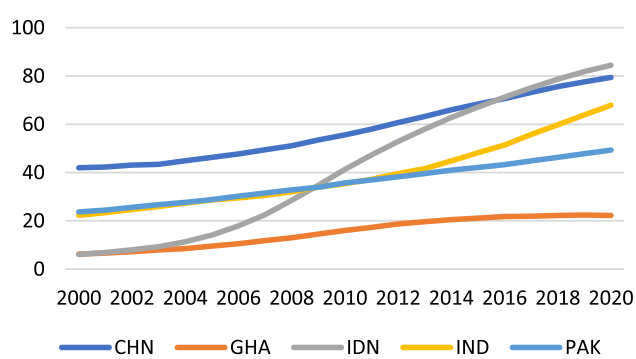

Fig. 1 Access to clean fuels and cooking technologies (% population). (Source: WHO Global Health Observatory, author's calculation)

exposure [6, 18, 19]. To the authors' knowledge, no study has been conducted to quantify the causal effects of the shift to clean cooking energy on mental health in Indonesia. Prior studies in Indonesia have focused on the effects of clean cooking energy on physical health outcomes, such as infant mortality and low birth weight [19], and health risks such as coughing, shortness of breath, fever, diarrhea, and eye pain [18]. Thus, to fill the research gap, this study will use longitudinal data to explore the causal impact of clean cooking energy transition on mental health in Indonesia.

This study makes three contributions to the existing literature. First, this paper examines the impact of household air pollution resulting from cooking with solid fuel on mental health, particularly depression symptoms. This is a significant health concern not previously investigated in developing countries. This study is believed to be the first attempt to examine the causal effects of transitioning to clean cooking energy on mental health in Indonesia through the utilization of the quasi-experimental methodology. Second, we examine the distinct effects of transitioning to different cooking energy sources on mental health in various regional contexts (rural and urban) and among different genders (women and men). This coincides with the fact that clean energy transition in Indonesia primarily focuses on urban regions, where the degree of stress experienced by residents tends to be higher compared to rural areas. In terms of gender, this research will determine whether women, who conduct more activities in the kitchen, will have a different impact than men. Third, this study uses longitudinal data and applies a combination of quasi-experiment estimation techniques, namely propensity score matching and difference-in-differences (PSM-DID) to reduce estimation bias caused by endogeneity arising from the selection mechanism. The DID approach enhances the visibility of the causal impact of the cooking energy transition on mental health. The evidence regarding the health effects of cooking with solid fuels is largely affected by endogeneity issues, as it predominantly relies on observational studies or cross-sectional data.

The subsequent sections of this paper are organized in the following order. Section 2 provides an overview of the existing literature. Section 3 provides comprehensive information, encompassing data sources and variables. Section 4 outlines the process of establishing treatment and control groups based on the samples. Section 5 outlines the estimation approach, which involves utilizing propensity score matching and difference in difference (DID) estimation. Section 6 covers several aspects including descriptive statistics, key findings, the heterogeneity of the impact, potential channeling, and robustness checks. Section 7 of the study discusses the final findings and offers recommendations for policy implementation.

2 Literature review

In their comprehensive study, Smith and Pillarisetti [20] provide a detailed analysis of the various paths through which the utilization of polluting fuels for residential purposes might have detrimental effects on human health. The environmental health pathway commences with the combustion of biomass fuels, leading to the subsequent emission of pollutants. This process is followed by the dispersion and accumulation of pollutants in the surrounding environment, as well as their gradual buildup within the human body due to regular exposure. Ultimately, these events culminate in various health repercussions. Grossman's [21] model, a notable theoretical framework for examining the generation of health capital, provides more insights into the underlying mechanism by which Household Air Pollution (HAP) impacts population health. Given an exogenously determined initial stock of health capital, the limitations on investing in health capital can be described by Eq. (1):

$$\Delta H_{t+1} = H_{t+1} - H_t = I_t - \delta_t H_t \tag{1}$$

If we define the stock of population health capital as H, then the difference between period t and period $t+1$ is denoted by ΔH_{t+1}. The health capital investment flow I_t for time period t. The rate of decline of health-related investments over time, denoted by δ_t. It has been established that the current level of HAP is positively connected with the depreciation rate of population health capital [22]. Higher HAP will result in quicker depreciation of population health capital, given a constant stock of health capital. In contrast, reduced HAP will result in slower depreciation of population health capital [23]. The health capital stock of people who live in places with inferior environmental circumstances tends to depreciate more rapidly.

The majority of the world's population typically allocates approximately 80–90% of their time to indoor environments characterized by limited spatial dimensions and restricted air circulation [24]. Du et al. [25] performed a comprehensive meta-analysis encompassing 92 articles pertaining to indoor air pollution in China. The findings of their research revealed noteworthy upwards trends in the concentrations of air pollutants in outdoor environments, bedrooms, and kitchens.

Yu et al. [24] study showed that indoor concentrations of these pollutants were generally two to five times greater than outside values. The detrimental effects of these hazardous air pollutants (HAPs) have the potential to shift the prevailing notion that homes are safe and comfortable environments, transforming them into one of the most perilous settings for human health.

A pollutant's effect on mental health might arise either directly, when the pollutant's biochemical mechanisms on neural tissue are disrupted, or indirectly when the pollutant causes chronic stress. A major risk factor for neurodegenerative disorders and neurodevelopmental disorders such as autism is fine particulate matter with a diameter of 2.5 m (PM2.5). According to the results of the study by Ventriglio et al. [26], PM2.5 may be a risk factor for depression. However, carbon monoxide (CO), another major air pollutant, has been shown to impair adult memory, learning, attention, and concentration by interfering with the development and function of the central nervous system [27].

Banerjee et al. [28] conducted an analysis to examine the effects of elevated concentrations of indoor air pollution, specifically long-term exposure to PM10 and PM2.5, on the severity of depression. The study focused on two groups: an experimental group consisting of 952 women in India who regularly utilized biomass stoves for cooking water at home, and a control group comprising 804 women who regularly used gas stoves for the same purpose. The findings indicated that female individuals who utilized biomass stoves for culinary purposes had elevated levels of depressive symptoms and experienced modifications in serotoninergic activity within the brain. Serotonin is a neurochemical that has a significant function in modulating mood and promoting positive affect. Furthermore, Nduka and Jimoh [29] investigate the impact of cooking energy sources and indoor air pollution on the happiness, life satisfaction, physical, and mental health of women in Nigeria by measuring Carbon Monoxide (CO) levels in participants' blood. The study finds that clean cooking energy usage is associated with a significant reduction in mental health problems among women.

Earlier studies by [16, 17, 30] in China indicated that the shift to clean cooking energy influences the decrease in Center for Epidemiological Studies Depression (CESD) scores and the likelihood of depression. Subsequently, Chen et al. [31] investigated how the utilization of clean fuel affects the health, chronic pain, and social engagement of the elderly population in China, providing insights into the mechanisms identified in their study.

The connection between the shift to clean cooking energy and mental health can be analyzed through two distinct mechanisms. The initial step involves decreasing indoor pollution, which leads to enhanced physical health. This improvement, in turn, boosts an individual's productivity and helps sustain their mental well-being. Additionally, the shift to clean cooking energy has the potential to reduce cooking time, thereby increasing leisure time. This increase in leisure can foster greater social participation, ultimately contributing to enhanced mental health. Evidence from Ma et al. [15] shows that switching to cleaner cooking fuels can have a positive effect on people's happiness and overall sense of contentment with life. The utilization of clean cooking energy enables individuals to optimize their time and energy spending in the cooking domain, so affording them additional opportunities to engage in various pursuits, including social activities, which have the potential to enhance their overall well-being and happiness. The preceding analysis of the literature allows us to formulate a research hypothesis: switching to clean cooking energy will improve mental health (in terms of lower CESD scores, lower rates of depression, and greater levels of happiness). The conceptual framework and hypothesis of the relationship between clean cooking energy transition and mental health are illustrated in Fig. 2.

3 Methodology

3.1 Data

The dataset utilized in this study is panel data obtained from the Indonesian Family Life Survey (ILFS). IFLS is a longitudinal study undertaken by the Rand Corporation, designed to be comprehensive and nationally representative (representing 83% of the Indonesian population) in scope. The IFLS produces detailed information at the individual, household, and community levels covering socioeconomics, demographic conditions, and health. It consists of five waves, but this study only uses the last two waves, IFLS 4 (in 2007) and IFLS 5 (in 2014), because both waves consistently include 10 CESD items, which are measures of depressive symptoms). Unfortunately, the previous three waves of the survey did not provide information related to 10-item CESD, which is important to define our main outcome variable (depressive symptoms). However, it is contended that the two survey waves employed in this research adequately capture the trajectory of the transition from kerosene to LPG for cooking in Indonesia, which began in 2007, and experienced significant growth from 2009 through 2011 [32]. It is possible to see a significant increase in households from using dirty fuel (kerosene,

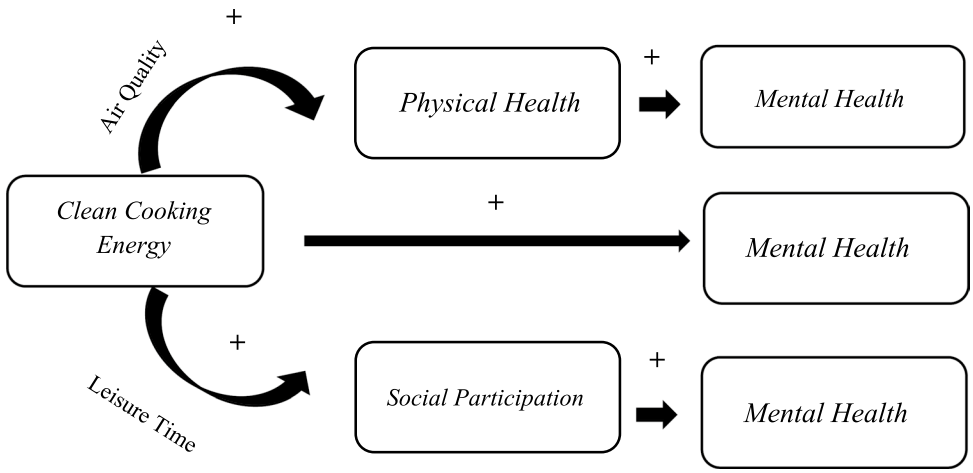

Fig. 2 Conceptual framework (Source: Author's synthesis)

firewood, charcoal, coal) to clean fuel (LPG) in the year 2014. In addition, this two-period panel data is valid for the empirical strategy used in this study.

This research employs three mental health indicators, specifically the CESD score, depression, and happiness. The CESD score is calculated by summing the respondent's answer values (ranging from 0 to 3) for a set of 10 question items, resulting in a possible range of scores from 0 to 30. The identification of depression is established by employing a CESD cut-off score of [33]. Individuals who obtain a CESD score exceeding 10 are considered to be experiencing symptoms of depression. Happiness is represented by a binary variable that distinguishes between respondents who are happy and those who are not happy.

3.2 Defining control and treatment group

The sample structure employed in this study begins with the transformation of IFLS 4 and 5 data into balanced panel data, with a total of 35,384 observations consisting of 17,692 individuals who were re-observed in two periods in 2007 and 2014. In order to identify the treatment and control groups within the IFLS dataset, individuals are classified into four distinct categories according to their cooking fuel switching status subsequent to the implementation of the LPG conversion program: people who continue to use dirty cooking, people who switch from dirty to clean cooking, people who switch from clean to dirty cooking, and people who switch from clean to dirty cooking. This study employed a sample structure that encompassed individuals who did not utilize clean cooking fuels in the year 2007, while excluding those who had utilized clean fuels within that time frame. This strategy is applicable to the estimate method known as Difference-in-Differences (DID), in which the pre-treatment time is consistent across all samples. The 4732 individuals in the control group continued to use dirty cooking fuels during both times, while the 9977 individuals in the treatment group went from dirty cooking in 2007 to clean cooking in 2014. The sample structure does not include the 2983 (2905 plus 78) individuals who utilized clean cooking in 2007. The Sankey diagram in Fig. 3 depicts the flow of changes in cooking fuel user status in IFLS between 2007 and 2014.

Figure 5 indicates that the treatment group is predominantly located in urban areas; conversely, those who do not transition to clean cooking energy reside in rural regions. This is comprehensible for the following reasons: the Indonesian government initiated LPG conversion, prioritizing big cities to enhance the existing LPG infrastructure. The availability and affordability of clean cooking energy in rural regions pose considerable obstacles. Hartono, Hastuti [34] assessed the accessibility and affordability of contemporary energy consumption in Indonesia, revealing that rural and low-income households require enhanced access.

3.3 Empirical strategy

An observational study aimed at making causal inferences, which utilizes data obtained via survey designs, is susceptible to bias due to the lack of randomization [35]. In addition, the execution of the strategy aimed at transitioning to clean cooking energy is not wholly random. Self-selection determines households' cooking fuel choices so selection bias is possible [17, 36]. It is imperative to mitigate the influence of selection bias in the initial sample. The

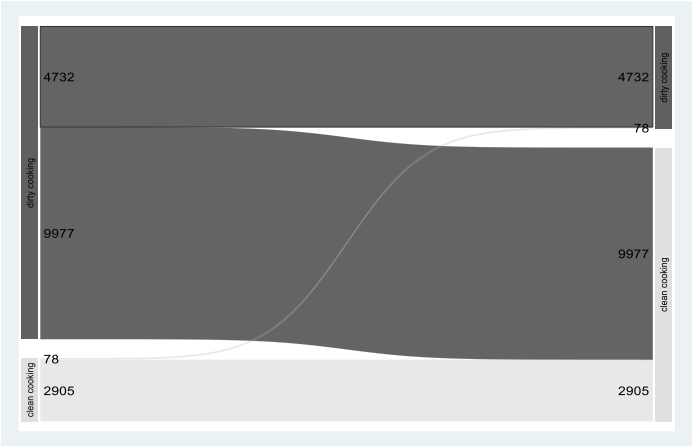

Fig. 3 Sankey graph of changes in cooking fuel users. (Source: IFLS 2007 & 2014, author's calculations)

cooking transition's effect on net mental health may be prejudiced if this selection bias is not rectified. Therefore, the (Propensity Score Matching) PSM approach is utilized. PSM is a statistical technique that relies on the counterfactual inference model. It is commonly employed to address the challenges posed by non-random data. The literature commonly uses a comparable methodology of matching to reduce selection bias [17, 36–38]. PSM matches samples from the control group (people who used non clean cooking energy (CCE) in both periods) with samples from the treatment group (people who switched from non-CCE to CCE in the post-treatment period) using a propensity score. The probability of a sample being in the treatment group with an observable variable X_i is the propensity score $p(X_i)$. First, estimate $p(X_i)$ with probit, and then predict the probability of selection using Eq. (2):

$$p(X_i) = \Pr(T_i = 1|X_i) = F(\widehat{\theta}X_i) \tag{2}$$

The objective of the matching strategy is to establish a control group that exhibits similar characteristics to the treatment group during the pretreatment period. Therefore, the resultant effect estimations will more accurately portray the consequences of the energy shift on mental health, rather than focusing on the distinct characteristics of the two groups. Furthermore, if the two groups—namely the treatment and control groups—formed through the matching approach, display a comparable trajectory of mental health development (referred to as a parallel trend) when the transition to clean cooking energy is absent, it becomes possible to estimate the Difference-in-Difference (DID). The DID approach is widely employed to assess the impact of exogenous shocks and establish causal relationships between economic variables. The DID model is employed in conducting a quasi-naturalistic trial [39]. Hence, the utilization of the PSM-DID approach, which combines Propensity Score Matching (PSM) with DID, is employed to enhance the precision of assessing the causal impact of the transition in cooking energy on mental health.

The control group consists of individuals who do not undergo cooking energy conversion during the specified time period. By subtracting the change in mental health of the control group from the difference in mental health before and after energy conversion in the treatment group, we obtain the net effect, also known as the average treatment effect. This calculation allows us to exclude the impact of unobservable time-variant variables. The canonical DID estimator is used to analyze the impact of the transition to clean cooking energy ($Treatment_i$) on mental health outcomes for individual i living on island r during time t (Y_{irt}^k). Superscript k denotes all three indicators of mental health (CESD, depression, and happiness). This estimator focuses on the comparison between the before and after periods, indicated by $POST_t$, and is characterized by the interest parameter $\boldsymbol{\beta}_3$.

$$Y_{irt}^k = \beta_0 + \beta_1 Treatment_i + \beta_2 POST_t + \boldsymbol{\beta}_3 Treatment_i x POST_t + \boldsymbol{X_{it}}\gamma + \delta_r + \varepsilon_{it} \tag{3}$$

The estimated value of $\boldsymbol{\beta}_3$ represents the quantification of the difference-in-difference in mental health-related outcome variables, specifically the CESD score, depression, and happiness, between individuals who transitioned from using dirty cooking energy to clean cooking energy and individuals who did not make this transition. This comparison is made by considering the time before and after the implementation of the clean cooking energy transition program, which involved converting to LPG. To account for potential confounding variables that may vary over time, the estimation in this study incorporates a collection of individual-level covariates denoted as $\boldsymbol{X}_{it}$. These covariates

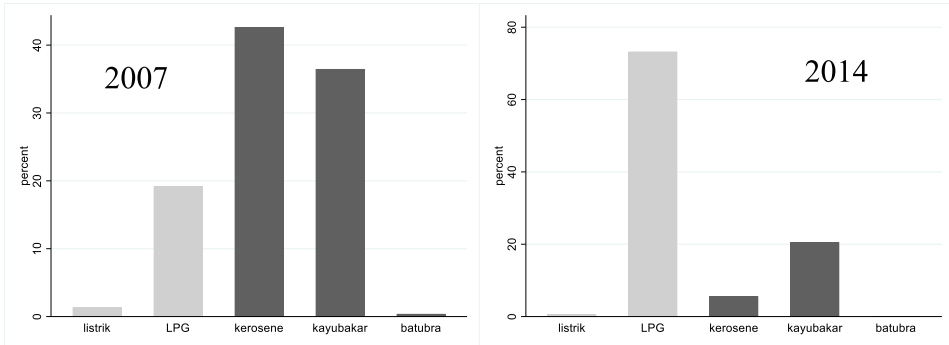

Fig. 4 The proportion of cooking fuel users in 2007 and 2014 (Source: IFLS 2007 and 2014, author's calculations)

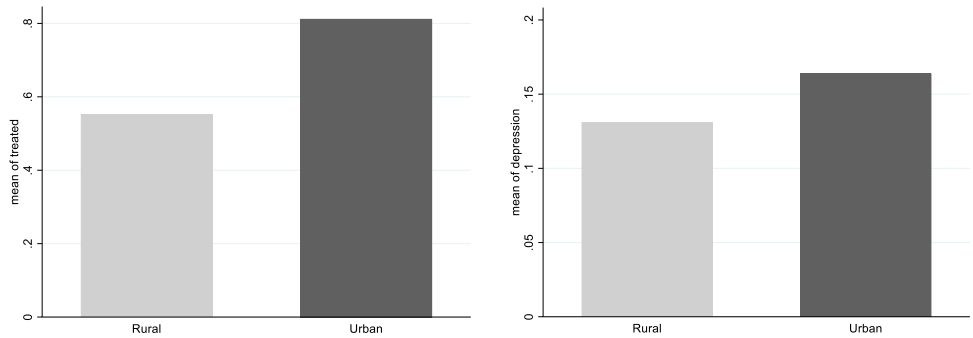

Fig. 5 The proportion of treated group and depression in rural and urban (Source: IFLS 2007 & 2014, author's calculations)

encompass factors such as gender, age, education level, employment status, smoking habits, ownership of insurance, possession of a house with a separate kitchen, availability of ventilation, access to electricity, possession of a refrigerator and television, per capita income, household fuel expenditure, and residence in either rural or urban areas. The use of these covariates follows previous research that has been conducted by [11, 17, 19].

While the specified set of covariates has been employed as matching covariates during the pre-treatment period (2007), it is plausible that there may have been alterations or fluctuations in the covariate values during the post-treatment period (2014), which could potentially affect the estimation outcomes of the CCE transition impact on β_3. Consequently, the estimate of the Difference-in-Differences (DID) model will continue to utilize the collection of covariates $\boldsymbol{X}_{it}$. The estimation results for the DID analysis, both with and without the control variables, are presented in Appendix 4. This analysis additionally incorporates island dummies δ_r to mitigate potential bias arising from macro aggregate shocks and varying degrees of development across the Indonesian archipelago.

4 Results

4.1 Descriptive statistic

Figure 4 shows that in 2007, the most common cooking fuel used by families in Indonesia was kerosene (43%), followed by firewood (36%). In 2014, the share of households utilizing CCE increased significantly to 79%, while non-CCE declined to 26%. In 2014, LPG was the most commonly used cooking fuel (73%). This is consistent with the Indonesian government's successful kerosene-to-LPG conversion program, which ran from 2009 to 2011.

The overall CESD score varied from 0 to 28, with a mean of 5.056. The dependability of the CESD score of the ten IFLS question items was assessed using Winstep 3.73.0 and found to be quite good, with a Cronbach Alpha close to 0.70 (the results of the analysis are shown in the Appendix 1). The CESD score is interpreted to mean that the greater the CESD score, the worse the depressed symptoms. The Indonesian government began the LPG conversion initiative in big cities in Java [32]. Figure 5 depicts a comparison of the proportion of the treated group and depression by area of residence in rural and urban areas. It can be seen that urban areas have a higher proportion of CCE transitions than rural. Individuals living in urban areas have a higher likelihood of depression than those living in rural areas. Understandably, stress levels

in urban are higher than in rural due to various factors such as population density, traffic congestion, work pressure, and outdoor air pollution.

4.2 Propensity score matching (PSM)

Kernel matching, nearest neighbor matching, and radius matching are frequently used in propensity score matching [17]. In the pretreatment phase, the probit probability model and Epanechnikov Kernel matching method with a bandwidth of 0.06 are used to evaluate the propensity score or conditional likelihood of clean cooking energy transfer. The 0.06 is the average optimal bandwidth which produces smoother fits [37, 38]. According to the matching results, nine observations in the treated group are off-support. Figure 6 depicts the propensity score graph produced by PSM. The additional estimation will rule out off-support observations. As a result, this study makes use of a total of 29,400 eligible observations. This figure is based on 14,700 people who were re-observed throughout two periods, pretreatment (2007) and posttreatment (2014). Individuals are separated into two groups: 4732 in the control group and 9968 in the treatment group. The descriptive statistics for the dependent and independent variables utilized in this investigation are shown in Table 1.

Table 2 shows the CCE transition selection results. The pretreatment probit estimation results show that people with higher per capita income, higher levels of education, a married or previously married status, possession of health insurance, access to electricity, ownership of a refrigerator and a television, as well as residing in urban areas, exhibit a higher propensity to adopt clean energy for cooking, which is consistent with previous studies that found that socioeconomic variables are important [40, 41]. The study by Tampakis et al. [42] emphasizes the critical role of education in facilitating a successful transition to clean energy. Older adults were also more likely to use solid fuels than clean cooking energy, possibly because they had a harder time adapting to new things and stoves.

Then, the quality of the match is discussed. Following the matching process, assess the distributions of covariates in both the treatment and control groups. If PSM is effective, these distributions ought to align closely, suggesting that the covariates are balanced and the strong ignorability assumption is more credible. Figure 7a shows how the p-score distributions were spread out before they were matched. Overall, the p-score for the treatment group is much higher than that for the control group. Figure 7b shows the p-score distribution diagram after matching. By looking at the two numbers, it was clear that PSM fixed the difference in scores between the two groups. The next step is to perform a balance test (Appendix 2). The balancing test findings in Appendix 2 demonstrate that most variables have a p-value greater than 0.05, suggesting that there is no significant difference between the treatment and control groups after matching. However, some variables (separation kitchen, married status, and ln per capita income) have a significant p-value ($p < 0.05$). Nevertheless, we incorporated all control variables in our estimation. We compared the two estimation outcomes (containing and eliminating variables that have met the balancing properties). The calculation utilizing the entire set of variables results in a small shift in values (significance remains). Our research indicates that because the

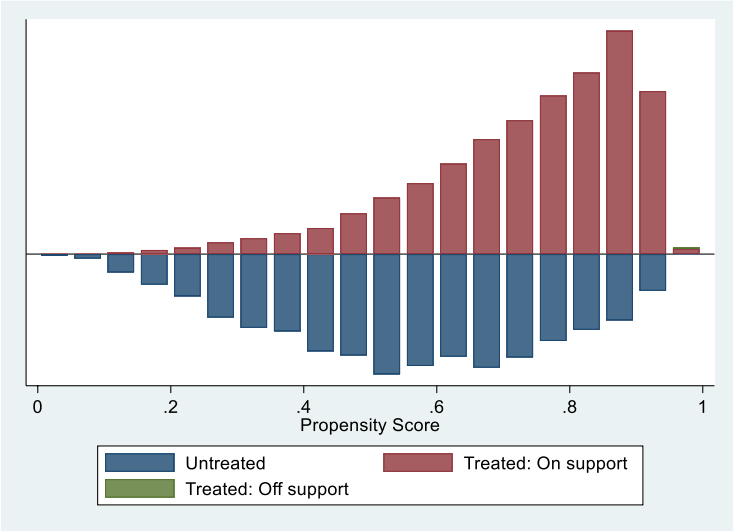

Fig. 6 Propensity score graph. (Source: IFLS 2007 & 2014, author's calculations)

Table 1 Descriptive statistics of research variables

Variables	Operational definition	Control Group			Treatment Group		
		Obs	Mean	SD	Obs	Mean	SD
cesd	CESD score (0–30)	9464	5.356	4.521	19,936	5.077	4.308
depression	depression symptoms: 0 = no; 1 = yes	9464	0.153	0.360	19,936	0.144	0.351
happy	self-rated happiness: 0 = not happy; 1 = happy	9464	0.870	0.336	19,936	0.924	0.265
health status	health status: 0 = unwell; 1 = healthy	9464	0.802	0.399	19,936	0.824	0.381
social	social participation: 0 = no; 1 = yes	9464	0.753	0.431	19,936	0.736	0.441
female	gender: 0 = male; 1 = female	9464	0.542	0.498	19,936	0.549	0.498
age	age ranges between 14–101 years	9464	41.772	14.812	19,936	37.675	13.043
i.education	education level: 4 categories						
uneducated	1: did not pass elementary school (base)	9464	0.103	0.305	19,936	0.034	0.181
primary educ	2: graduated from elementary school	9464	0.514	0.500	19,936	0.369	0.483
secondary educ	3: graduate from high school	9464	0.341	0.474	19,936	0.511	0.500
tertiary educ	4: graduated from college	9464	0.041	0.199	19,936	0.085	0.280
i.marital_status	marital status: 3 categories						
unmarried	1: unmarried (base)	9464	0.115	0.318	19,936	0.133	0.340
married	2: married	9464	0.789	0.408	19,936	0.797	0.402
divorced	3: divorced/separated/widowed	9464	0.096	0.295	19,936	0.070	0.255
work	working status: 0 = not working; 1 = work	9464	0.771	0.420	19,936	0.708	0.455
smoking	smoking habit: 0 = no smoking; 1 = smoking	9464	0.383	0.486	19,936	0.350	0.477
has insurance	health insurance: 0 = do not have; 1 = has	9464	0.339	0.473	19,936	0.376	0.484
separated kitchen	kitchen separates from the main house: 0 = no; 1 = separate	9464	0.271	0.444	19,936	0.229	0.420
ventilation	ventilation in the house; 0 = none; 1 = exists	9464	0.834	0.372	19,936	0.857	0.350
electricity	electricity access: 0 = none; 1 = exists	9464	0.950	0.218	19,936	0.988	0.107
refrigerator	refrigerator ownership: 0 = do not have; 1 = has	9464	0.296	0.456	19,936	0.539	0.499
tv	television ownership: 0 = do not have; 1 = has	9464	0.725	0.446	19,936	0.897	0.305
ln per-capita income	log form of annual household per capita income (rupiah)	9464	15.507	1.165	19,936	15.985	1.176
ln fuels expenditure	log form of household per capita expenditure (rupiah)	9464	7.338	4.707	19,936	9.888	2.828
urban	area of residence: 0 = rural; 1 = urban	9464	0.283	0.451	19,936	0.581	0.493

Source: IFLS 2007 & 2014, author's calculations

matching is done in 2007, and the impact estimation employs 2007 and 2014 data, there is a change in the covariate values in both periods, which influences the estimation findings.

The assumption of strong ignorability in propensity score matching is essential for ensuring its validity. This assumption in PSM can be seen from the c-statistic and pseudo-R^2 values to evaluate the model's performance and quality in estimating treatment effects. The c-statistic, commonly referred to as the area under the ROC curve (AUC), evaluates the model's discriminatory ability. It evaluates the model's ability to differentiate between two groups (treatment versus control) utilizing the estimated probabilities generated by the model. A c-statistic value of 0.5 signifies a lack of discrimination, whereas a value of 1.0 denotes flawless discrimination. Within the framework of Propensity Score Matching (PSM), elevated c-statistic values signify that the propensity score model effectively forecasts treatment allocation based on observable factors [43]. We have computed the c-statistic, which statistically assesses the discriminating capability of the propensity score model. The c-statistic for the propensity model in this study was 0.75. The propensity score model effectively estimates treatment assignment based on observable data.

In Appendix 2, it is evident that the pseudo-R2 value has significantly decreased after matching, from 0.144 to 0.002. The notable reduction in the pseudo-R^2 following matching suggests an improved balance of covariates between the treatment and control groups, indicating that the matching process has successfully addressed confounding variables. The reduction signifies enhanced comparability among the matched samples, thereby augmenting the validity of the study's causal findings. This aligns with the research by [44], which examined the impact of poverty reduction policies on health services utilization in China. Consequently, we can conclude that the PSM employed in this study met the "strong ignorability" condition.

Table 2 Selection on the clean cooking energy transition (Probit)

Variable	2007	
	Coefficient	Standard Error
female	−0.054**	0.025
age	−0.012***	0.001
primary	0.140***	0.052
secondary	0.279***	0.057
tertiary	0.338***	0.076
married	0.217***	0.036
divorced	0.258***	0.062
work	−0.152***	0.028
insurance	0.060**	0.028
separated kitchen	−0.032	0.025
ventilation	0.006	0.031
electricity	0.409***	0.060
refrigerator	0.221***	0.027
tv	0.355***	0.029
ln per-capita income	0.117***	0.012
ln fuels expenditure	0.052***	0.003
urban	0.509***	0.025
constant	−2.534***	0.196
N	14,709	
Pseudo-R^2	0.144	

*$p<0.1$, **$p<0.05$, ***$p<0.01$ (Source: IFLS 2007, author's calculations)

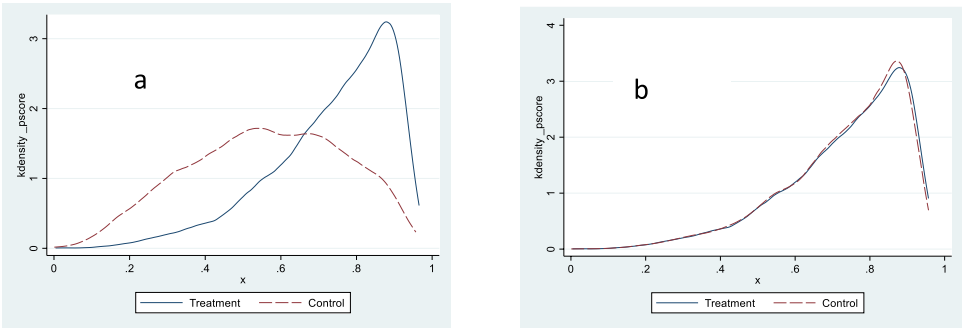

Fig. 7 Propensity Score Distributions (**a** before matching, **b** post matching). (Source: IFLS 2007 & 2014, author's calculations)

4.3 The impact of clean cooking energy transition to mental health

Table 3 displays the major regression results used to calculate the impact of the clean cooking energy transition on mental health. Columns (1), (2), and (3) illustrate the influence of cooking energy transition on CESD score, depression probability, and happiness without using matching weights (unweighted), but columns (4), (5), and (6) do (weighted). In this study, the variable of interest $\beta_3 = \beta_{did}$ is the interaction between dummy treatment and dummy time (post), commonly known as the average treatment effect.

The β_{did} coefficient in column (1) shows a negative value of 0.326 significant at the 1% level, indicating that the treatment group has an average CESD score of 0.326 lower than the control group after the CCE conversion program. While utilizing weighted matching, the β_{did} coefficient in column (4) shifts to negative 0.361 but stays significant at the 1% level, indicating that the treatment group had a 0.361 lower average CESD score than the control group following the CCE conversion program. This demonstrates that the coefficient in column (1) is still undervalued (negative bias) since the matching weight was not taken into account. Therefore, we can conclude that the clean cooking energy shift contributes to the reduction of individual CESD scores.

Table 3 Estimation results of the impact of clean cooking energy transition on mental health

	Unweighted			Weighted		
	(1)	(2)	(3)	(4)	(5)	(6)
	CESD	Depression	Happiness	CESD	Depression	Happiness
treatment = 1	−0.096	−0.001	0.012**	−0.065	−0.000	0.017***
	(0.067)	(0.005)	(0.006)	(0.085)	(0.006)	(0.006)
post = 1	2.946***	0.208***	−0.042***	2.988***	0.207***	−0.027***
	(0.094)	(0.008)	(0.007)	(0.131)	(0.011)	(0.008)
treatment = 1 # post = 1	**−0.326*** **	**−0.031*** **	**0.019** **	**−0.361*** **	**−0.029** **	**0.005**
	(0.107)	(0.009)	(0.008)	(0.139)	(0.012)	(0.009)
control variables	Yes	Yes	Yes	Yes	Yes	Yes
i.island	Yes	Yes	Yes	Yes	Yes	Yes
Constant	8.391***	0.276***	0.572***	9.236***	0.330***	0.546***
	(0.459)	(0.036)	(0.035)	(0.586)	(0.047)	(0.041)
Observations	29,400	29,400	29,400	29,400	29,400	29,400
R^2	0.094	0.068	0.051	0.100	0.071	0.046

The bolded coefficient represents the DID estimator (betadid)

Standard errors in parentheses; $^{*}p < .1$, $^{**}p < .05$, $^{***}p < .01$

The estimation findings for the outcome variable depression in column (2) coefficient β_{did} reveal a negative value of 0.031 significant at the 1% level, indicating that following the CCE conversion program, the treatment group had a 3.1% lower likelihood of depression than the control group. Because the matching weights were not taken into account, this result still has a favorable bias (is overstated). When the matching weights in column (5) are used, the coefficient β_{did} declines to a significant negative value of 0.029 at the 5% level. This suggests that following the CCE conversion program, the treatment group had a 2.9% reduced risk of depression compared with the control group. As a result, the CCE transition can be argued to have an effect on lowering the likelihood of depression.

Furthermore, without matching weights, the β_{did} coefficient has a positive value of 0.019 and is significant at the 5% level in the estimated results with the outcome variable happiness in columns (3) and (6). This suggests that following the CCE conversion, the treatment group has a 1.9% higher chance of being happy than the control group. However, when weighted matching is used, the coefficient drops to 0.005 and is no longer significant. As a result, the CCE transition has no effect on the likelihood of individual happiness.

The findings derived from this investigation are consistent with prior research that has similarly examined the impact of transitioning to clean cooking energy on mental well-being. In a series of case studies conducted in China [17, 30, 45] observed consistent findings indicating that the utilization of clean cooking energy is associated with a reduction in CESD scores and a decreased likelihood of experiencing depression. When examining the amount of the CESD score reduction, it is observed that the impact size of 0.361 in the present study is somewhat larger than the effect size of 0.33 reported by [11]. Similarly, Liu et al. [17] discovered a decrease in the likelihood of depression with a magnitude of 0.019 (equivalent to 1.9%). In contrast, our study observed a greater impact size of 0.029 (equivalent to 2.9%). This implies that the impact of transitioning to clean cooking energy on mental health holds greater significance in Indonesia—a lower-middle-income country—compared to a high-middle-income country.

Furthermore, this study assessed the economic significance of the findings by employing the elasticity of the effect coefficient derived from statistical analysis, relative to the mean value of the outcome variable in the control group during the pretreatment period. The findings indicate that the transition to clean cooking energy had a significant effect on reducing the CESD score by 8.9% (calculated as 0.361 divided by 4.059). Additionally, it was observed that this shift led to a substantial decrease of 48.3% in the likelihood of feeling depression (calculated as 0.29 divided by 0.060). The economic significance of the clean cooking energy shift in Indonesia is seen in its positive influence on mental health.

The use of classical Difference-in-Differences implies adherence to the parallel trend and common shock assumptions. Due to the unavailability of the 10 items-CESD data in IFLS before to 2007, the examination of parallel trend and common shock assumptions cannot be conducted directly in this study. We consider this a shortcoming of the study. Nonetheless, referring to studies that employ analogous estimation methodologies and similarly do not directly evaluate both assumptions, such as Gitaharie et al. (2022) in their investigation of ex-ante moral hazard in health insurance in Indonesia and Luo et al. (2023) in their assessment of the Low Carbon City Pilot (LCCP) on corporate environmental expenditure in

China, the authors contend that the magnitude of the impact derived from this estimation remains credible. We implemented matching in the pre-treatment phase, accompanied by an extensive array of variables, prior to implementing the DID estimate. The matching approach aims to provide a comparable control group for those switching to clean cooking energy. Consequently, our impact estimates will more precisely reflect the impacts of the shift to clean cooking energy on mental health, rather than differences in characteristics between the two groups. Incorporate control variables in our study to account for any confounding influences. This may mitigate the impact of observable confounding variables.

4.4 Potential channeling analysis

Table 4 shows the estimation findings of the mediation study. Column (1) reveals that the treatment group has a 2.6% greater chance of improving physical health than the control group following the conversion program. In other words, cooking energy transfer significantly improves physical health. Column (3) further demonstrates that the clean cooking energy transition increases the likelihood of individuals participating socially by 3.8%. This is consistent with the findings of [17] in China. Furthermore, to estimate the extent of change in β_{did}, the physical health status variable is regressed as a control variable. When compared to column (5) before the mediator, the magnitude of the coefficient of the variable of interest in columns (2) and (4) decreases after including the mediating variable. This demonstrates that household air pollution has an indirect impact on mental health via physical health and social participation.

4.5 Heterogeneity analysis

The impact of the transition to clean cooking energy on mental health could exhibit variability based on the specific subsample under consideration. Individuals who allocate a substantial portion of their time in the cooking domain, such as the kitchen, may exhibit a heightened receptiveness towards the advantageous outcomes associated with the change in cooking energy. Consequently, this study incorporates a heterogeneity analysis to examine the impact of the transition in cooking energy on two distinct subsamples: gender-based groups (female and male) and location-based groups (urban and rural), to conduct a comprehensive analysis.

Table 5 demonstrates that the influence of the cooking energy transition on CESD scores and the likelihood of experiencing depression is stronger in the female subsample than in the male subsample. This is consistent with previous findings [17, 36] that women will be more sensitive to the impact of the cooking energy transition because women are typically responsible for cooking and doing household chores, exposing them to household air pollution for a longer period than men. In terms of the subsample of urban versus rural inhabitants, it appears that the influence of the

Table 4 Estimation results of mediation analysis (potential channeling)

	(1) Health_Status	(2) Depression	(3) Social	(4) Depression	(5) Depression
treatment = 1	−0.010	−0.003	−0.010	−0.001	−0.001
	(0.006)	(0.005)	(0.008)	(0.005)	(0.005)
post = 1	−0.112***	0.192***	0.012	0.208***	0.208***
	(0.009)	(0.008)	(0.009)	(0.008)	(0.008)
treatment = 1 # post = 1	**0.026*** **	**−0.027*** **	**0.038*** **	**−0.030*** **	**−0.031*** **
	(0.010)	(0.009)	(0.011)	(0.009)	(0.009)
health status		−0.145***			
		(0.006)			
social participation				−0.003	
				(0.005)	
control variables	Yes	Yes	Yes	Yes	Yes
i.island	Yes	Yes	Yes	Yes	Yes
constant	0.828***	0.396***	0.405***	0.275***	0.276***
	(0.040)	(0.036)	(0.043)	(0.036)	(0.036)
Observations	29,400	29,400	29,400	29,400	29,400
R^2	0.058	0.092	0.080	0.068	0.068

The bolded coefficient represents the DID estimator (betadid)

Standard errors in parentheses, $^{*}p < .1$, $^{**}p < .05$, $^{***}p < .01$

Table 5 Estimation results of heterogeneity analysis

		(1) CESD	(2) Depression	(3) Happiness
Gender	Women	**−0.412*****	**−0.035*****	**0.031*****
		(0.147)	(0.012)	(0.010)
	Obs	16,080	16,080	16,080
	R^2	0.095	0.071	0.052
	Men	**−0.228**	**−0.027****	**0.005**
		(0.155)	(0.013)	(0.012)
	Obs	13,320	13,320	13,320
	R^2	0.099	0.070	0.058
Region	Urban	**−0.497*****	**−0.052*****	**0.039*****
		(0.183)	(0.015)	(0.013)
	Obs	14,274	14,274	14,274
	R^2	0.094	0.065	0.049
	Rural	**−0.277****	**−0.024****	**0.008**
		(0.141)	(0.011)	(0.010)
	Obs	15,126	15,126	15,126
	R^2	0.095	0.072	0.053
Control Variables		Yes	Yes	Yes
i.island		Yes	Yes	Yes

Standard errors in parentheses, $^{*}p < .1$, $^{**}p < .05$, $^{***}p < .01$

clean cooking energy shift on mental health is more pronounced in urban residents. This is an intriguing finding for the Indonesian case study because, in addition to higher stress levels in cities, the Indonesian government began CCE conversion by targeting urban people first. It was clear that the CCE transformation has a larger impact on improving mental health in urban areas.

4.6 Robustness check

As a robustness check, various CESD cut-off scores, specifically CESD scores 12 and 15, are used to identify the depression variable [9, 16, 17]. The robustness check estimation result, shown in Table 6, indicates that the β_{did} coefficients in columns (1), (2), and (3) have the same direction and significance level. This ensures that the estimates of the influence of the clean cooking energy transition on mental health in this study are robust.

Table 6 Robustness check estimation results

	(1) Depression (cut-off CESD ≥ 10)	(2) Depression (cut-off CESD ≥ 12)	(3) Depression (cut-off CESD ≥ 15)
treatment = 1	−0.001	−0.002	−0.001
	(0.005)	(0.004)	(0.003)
after = 1	0.208***	0.153***	0.080***
	(0.008)	(0.007)	(0.005)
treatment = 1 # after = 1	**−0.031*****	**−0.026*****	**−0.023*****
	(0.009)	(0.007)	(0.006)
i.island	Yes	Yes	Yes
control variables	Yes	Yes	Yes
Constant	0.276***	0.213***	0.089***
	(0.036)	(0.032)	(0.023)
Observations	29,400	29,400	29,400
R^2	0.068	0.052	0.027

Standard errors in parentheses, $^{*}p < .1$, $^{**}p < .05$, $^{***}p < .01$

Furthermore, this study incorporates additional robustness checks by employing other matching approaches, such as closest neighbor 5 and caliper radius 0.01, in addition to utilizing various depression cut-off points. The analysis in Appendix 5 demonstrates that the use of the three matching approaches yields coefficients of interest that exhibit consistent directionality, comparable magnitudes, and equivalent levels of statistical significance. This enhances the validity of the robustness analysis conducted to assess the findings pertaining to the estimation of the effects of the transition to clean cooking energy on mental health.

5 Conclusion and policy implications

Using the Propensity Score Matching and Difference-in-Difference (PSM-DID) estimation method, this study intends to assess the impact of the clean cooking energy transition on mental health in Indonesia. This study establishes a significant correlation between the adoption of clean cooking energy and enhanced mental health, as seen by decreased CESD scores and lowered risk of depression. The results indicate that this positive impact is especially significant among women and those who live in urban areas. Switching to clean cooking energy can have a direct or indirect positive impact on mental health via the mediators of increased physical health status and social participation. The findings indicate that clean cooking energy initiatives may effectively enhance mental well-being in populations, especially among vulnerable demographics. The integration of mental health into wider energy and development policies is essential. Clean cooking energy interventions ought to be evaluated not just for their environmental and health advantages, but also for their capacity to enhance mental well-being.

The results of this study provide important insights that can inform policy recommendations for the government. The Indonesian government is justified in initiating a clean cooking energy transition program through an LPG conversion initiative in urban areas, given that urban residents experience higher stress levels compared to their rural counterparts. The evidence indicates that the clean cooking energy transition significantly influences the reduction of CESD and depression scores while enhancing happiness more effectively in urban settings compared to rural ones. Clean cooking energy conversion in rural areas, however, continues to be lower than in urban settings, even with the less stress observed in communities. To address mental health challenges, the government might explore the option of enhancing access to clean cooking energy in rural areas.

Given the significant implications for women, it is essential that policies tackle the gendered division of labor associated with cooking and fuel collection. This could involve enabling women to participate in decision-making about household energy options and enhancing their access to clean cooking technologies. Furthermore, it is essential for public awareness campaigns to emphasize the mental health advantages of clean cooking energy in addition to the more frequently highlighted physical and environmental benefits. This has the potential to enhance public backing for clean energy transitions and promote the adoption of cleaner cooking practices.

Although our findings provide promising insights into the effects of transitioning to clean cooking energy on mental health, this study did not account for fuel price as a variable affecting the selection of cooking fuel due to data limitations. This inquiry will be postponed for future investigation. Given the lack of comprehensive studies and the importance of this subject, we suggest that more attention should be directed towards exploring the relationship between cooking energy and mental health. Additional investigation is essential to comprehend the elements influencing the urban–rural gap in the effects of clean cooking energy on mental well-being. Policies must tackle the distinct obstacles encountered by rural communities in obtaining clean cooking technologies and guarantee an equal distribution of resources.

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Data availability The datasets generated during and/or analyzed during the current study are available from the corresponding author upon reasonable request.

Declarations

Ethics approval Not applicable.

Consent for publication Not applicable.

Consent to participate Not applicable.

Competing interests The authors declare no competing interests.

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Appendices

Appendix 1. Summary of measurement error and reliability of CESD score

Summary of 29019 measured (extreme and non-extreme) Person								IFLS 2007
	Total			Model	Infit		Outfit	
	Score	Count	Measure	Error	MNSQ	ZSTD	MNSQ	ZSTD
Mean	4.1	10.0	−1.53	0.63				
S.D	3.6	0.0	0.81	0.42				
Max	30	10.0	3.44	1.79				
Min	0.0	10.0	−3.18	0.30	0.12	−2.9	0.12	−2.1
Real RMSE	0.78	True SD	0.23	Separation	0.29	Person reliability		0.08
Model RMSE	0.76	True SD	0.29	Separation	0.38	Person reliability		0.12
S.E. of Person Mean		0.00						
Person raw score-to-measure Correlation = 0.88								
Cronbach Alpha (KR-20) Person RAW score "test" reliability = 0.54								
Summary of 31447 measured (extreme and non-extreme) Person								IFLS 2014
	Total			Model	Infit		Outfit	
	Score	Count	Measure	Error	MNSQ	ZSTD	MNSQ	ZSTD
Mean	6.4	10.0	−1.19	0.53				
S.D	4.8	0.0	0.90	0.37				
Max	30	10.0	3.91	1.83				
Min	0.0	10.0	−3.50	0.30	0.05	−6.1	0.05	−6.0
Real RMSE	0.67	True SD	0.60	Separation	0.89	Person reliability		0.44
Model RMSE	0.64	True SD	0.63	Separation	0.97	Person reliability		0.49
S.E. of person mean		0.01						
Person raw score-to-measure correlation = 0.89								

Summary of 31447 measured (extreme and non-extreme) Person								IFLS 2014
	Total			Model	Infit		Outfit	
	Score	Count	Measure	Error	MNSQ	ZSTD	MNSQ	ZSTD
Cronbach Alpha (KR-20) Person RAW Score "Test" Reliability = 0.69								

Appendix 2. Balance properties pre and post matching

Variable	Unmatched Matched	Mean Treated	Mean Control	%bias	%reduct |bias|	t-test t	t-test p>|t|	V(T)/ V(C)
gender2007	U	.54906	.54227	1.4		0.77	0.439	.
	M	.54916	.55109	-0.4	71.5	-0.27	0.784	.
age2007	U	34.219	38.345	-30.5		-17.71	0.000	0.76*
	M	34.233	34.036	1.5	95.2	1.11	0.267	1.01
2.EDUC2007	U	.37466	.52071	-29.7		-16.92	0.000	.
	M	.375	.37822	-0.7	97.8	-0.47	0.639	.
3.EDUC2007	U	.5203	.34721	35.5		19.92	0.000	.
	M	.51986	.52108	-0.2	99.3	-0.17	0.864	.
4.EDUC2007	U	.07086	.0317	17.8		9.50	0.000	.
	M	.07093	.06762	1.5	91.6	0.92	0.358	.
2.marital2007	U	.75223	.76669	-3.4		-1.91	0.056	.
	M	.75251	.72636	6.1	-80.7	4.21	0.000	.
3.marital2007	U	.05352	.07713	-9.6		-5.59	0.000	.
	M	.05357	.06129	-3.1	67.3	-2.34	0.019	.
work2007	U	.67355	.75993	-19.3		-10.73	0.000	.
	M	.67416	.66766	1.4	92.5	0.98	0.329	.
asuransi2007	U	.25348	.2145	9.2		5.17	0.000	.
	M	.25341	.24201	2.7	70.8	1.86	0.062	.
dapur_pisah2007	U	.31172	.33305	-4.6		-2.59	0.009	.
	M	.3117	.29572	3.4	25.1	2.45	0.014	.
ventilasi2007	U	.84544	.82079	6.6		3.79	0.000	.
	M	.8453	.83579	2.6	61.4	1.84	0.066	.
listrik2007	U	.98076	.91674	29.3		18.76	0.000	.
	M	.98074	.9804	0.2	99.5	0.17	0.864	.
kulkas2007	U	.38288	.19548	42.2		23.12	0.000	.
	M	.38232	.38499	-0.6	98.6	-0.39	0.699	.
tv2007	U	.83753	.61961	50.5		30.11	0.000	.
	M	.83738	.83948	-0.5	99.0	-0.40	0.687	.
lninc2007	U	15.605	15.144	43.8		25.51	0.000	0.73*
	M	15.603	15.537	6.3	85.5	5.05	0.000	1.16*
lnexp2007	U	9.6924	7.9771	47.6		28.42	0.000	0.57*
	M	9.6909	9.6144	2.1	95.5	1.84	0.067	1.20*
urban2007	U	.52862	.24937	59.8		33.04	0.000	.
	M	.52819	.5195	1.9	96.9	1.23	0.219	.

* if variance ratio outside [0.96; 1.04] for U and [0.96; 1.04] for M

Sample	Ps R2	LR chi2	p>chi2	MeanBias	MedBias	B	R	%Var
Unmatched	0.144	2656.84	0.000	25.9	29.3	95.8*	0.78	100
Matched	0.002	61.09	0.000	2.1	1.5	11.1	1.12	67

* if B>25%, R outside [0.5; 2]

Appendix 3. Overall estimation of the impact of the clean cooking energy transition on mental health

	Unweighted			Weighted		
	(1)	(2)	(3)	(4)	(5)	(6)
	CESD	Depression	Happiness	CESD	Depression	Happiness
treatment = 1	−0.096	−0.001	0.012**	−0.065	−0.000	0.017***
	(0.067)	(0.005)	(0.006)	(0.085)	(0.006)	(0.006)
post = 1	2.946***	0.208***	−0.042***	2.988***	0.207***	−0.027***
	(0.094)	(0.008)	(0.007)	(0.131)	(0.011)	(0.008)
treatment = 1 # post = 1	**−0.326*** **	**−0.031*** **	**0.019** **	**−0.361*** **	**−0.029** **	**0.005**
	(0.107)	(0.009)	(0.008)	(0.139)	(0.012)	(0.009)
female = 1	0.516***	0.029***	−0.003	0.548***	0.030***	−0.006
	(0.076)	(0.006)	(0.005)	(0.109)	(0.009)	(0.007)
age, range 14–101	−0.024***	−0.002***	−0.002***	−0.022***	−0.002***	−0.002***
	(0.002)	(0.000)	(0.000)	(0.003)	(0.000)	(0.000)
primary	0.093	0.031***	0.011	0.195	0.039***	0.007
	(0.115)	(0.009)	(0.010)	(0.138)	(0.011)	(0.011)
secondary	−0.248**	0.010	0.035***	−0.209	0.014	0.032***
	(0.124)	(0.010)	(0.011)	(0.151)	(0.012)	(0.012)
tertiary	−0.330**	0.005	0.057***	−0.335*	0.010	0.058***
	(0.151)	(0.012)	(0.011)	(0.197)	(0.016)	(0.013)
married	−0.497***	−0.027***	0.049***	−0.560***	−0.034***	0.040***
	(0.083)	(0.007)	(0.006)	(0.121)	(0.010)	(0.007)
divorced	0.169	0.008	−0.041***	0.121	0.011	−0.063***
	(0.141)	(0.011)	(0.011)	(0.204)	(0.017)	(0.016)
work = 1	−0.114*	−0.002	0.016***	−0.075	−0.002	0.015***
	(0.060)	(0.005)	(0.004)	(0.085)	(0.007)	(0.006)
smoking = 1	0.664***	0.034***	−0.026***	0.670***	0.031***	−0.033***
	(0.077)	(0.006)	(0.005)	(0.108)	(0.009)	(0.007)
has insurance = 1	0.120**	0.013***	−0.009**	0.106	0.011*	−0.010**
	(0.054)	(0.005)	(0.004)	(0.076)	(0.006)	(0.005)
separated kitchen = 1	0.144**	0.010**	−0.013***	0.102	0.012*	−0.017***
	(0.058)	(0.005)	(0.004)	(0.084)	(0.007)	(0.006)
has ventilation = 1	−0.220***	−0.011*	0.023***	−0.294***	−0.016*	0.021***
	(0.071)	(0.006)	(0.005)	(0.105)	(0.009)	(0.007)
has electricity = 1	−0.207	−0.008	0.016	−0.116	0.005	0.017
	(0.153)	(0.012)	(0.014)	(0.194)	(0.016)	(0.017)
has refrigerator = 1	−0.100*	−0.008*	0.012***	−0.068	−0.004	0.006
	(0.058)	(0.005)	(0.004)	(0.080)	(0.007)	(0.005)
has TV = 1	−0.128*	0.005	0.035***	−0.056	0.010	0.028***
	(0.072)	(0.006)	(0.006)	(0.101)	(0.008)	(0.007)
ln income per capita	−0.210***	−0.012***	0.016***	−0.268***	−0.016***	0.019***
	(0.027)	(0.002)	(0.002)	(0.035)	(0.003)	(0.002)
ln HH fuels exp	0.012*	0.001*	0.002***	0.008	0.000	0.001*
	(0.007)	(0.001)	(0.001)	(0.010)	(0.001)	(0.001)
urban = 1	0.406***	0.025***	−0.010***	0.513***	0.032***	−0.009**
	(0.054)	(0.004)	(0.004)	(0.070)	(0.006)	(0.004)
Java	−0.237***	0.007	0.012***	−0.466***	−0.008	0.017***
	(0.063)	(0.005)	(0.004)	(0.088)	(0.007)	(0.006)
Bali, NTT, NTB	−0.005	0.015**	0.018***	−0.245*	−0.002	0.026***

	Unweighted			Weighted		
	(1)	(2)	(3)	(4)	(5)	(6)
	CESD	Depression	Happiness	CESD	Depression	Happiness
	(0.088)	(0.007)	(0.006)	(0.131)	(0.011)	(0.008)
Kalimantan	0.043	0.026***	0.060***	−0.218	0.015	0.064***
	(0.113)	(0.009)	(0.006)	(0.155)	(0.014)	(0.009)
Sulawesi	0.891***	0.079***	0.054***	0.581***	0.060***	0.051***
	(0.130)	(0.011)	(0.007)	(0.164)	(0.014)	(0.009)
Constant	8.391***	0.276***	0.572***	9.236***	0.330***	0.546***
	(0.459)	(0.036)	(0.035)	(0.586)	(0.047)	(0.041)
Observations	29,400	29,400	29,400	29,400	29,400	29,400
R^2	0.094	0.068	0.051	0.100	0.071	0.046

The bolded coefficient represents the DID estimator (betadid)
Standard errors in parentheses
* *p* .1, ** *p* .05, *** *p* .01

Appendix 4. Robustness check (with and without control variables)

	(1)	(2)	(3)	(4)	(5)	(6)
	CESD	CESD	Depression	Depression	Happiness	Happiness
treatment = 1	-0.127	−0.065	−0.002	−0.000	0.016***	0.017***
	(0.084)	(0.085)	(0.006)	(0.006)	(0.006)	(0.006)
post = 1	2.653***	2.988***	0.185***	0.207***	−0.024***	−0.027***
	(0.126)	(0.131)	(0.010)	(0.011)	(0.008)	(0.008)
treatment = 1 # post = 1	**−0.387*** **	**−0.361*** **	**−0.031*** **	**−0.029** **	**0.013**	**0.005**
	(0.139)	(0.139)	(0.012)	(0.012)	(0.009)	(0.009)
control variables	No	Yes	No	Yes	No	Yes
i.island	Yes	Yes	Yes	Yes	Yes	Yes
constant	4.071***	9.236***	0.069***	0.330***	0.914***	0.546***
	(0.076)	(0.586)	(0.005)	(0.047)	(0.005)	(0.041)
Observations	29,400	29,400	29,400	29,400	29,400	29,400
R^2	0.078	0.100	0.057	0.071	0.003	0.046

The bolded coefficient represents the DID estimator (betadid)
Standard errors in parentheses
* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Appendix 5. Robustness check uses different methods for matching

	Kernel Epan Bandwith(0.06)			Nearest Neighbor 5			Radius Caliper (0.01)		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	CESD	Depression	Happiness	CESD	Depression	Happiness	CESD	Depression	Happiness
treatment = 1	−0.065	−0.000	0.017***	−0.039	0.002	0.016**	−0.069	0.000	0.016***
	(0.085)	(0.006)	(0.006)	(0.091)	(0.006)	(0.007)	(0.088)	(0.006)	(0.006)
post = 1	2.988***	0.207***	−0.027***	2.992***	0.203***	−0.025***	2.968***	0.206***	−0.028***
	(0.131)	(0.011)	(0.008)	(0.140)	(0.012)	(0.009)	(0.135)	(0.011)	(0.008)
treatment = 1 # post = 1	**−0.361*** **	**−0.029** **	**0.005**	**−0.369** **	**−0.027** **	**0.002**	**−0.342** **	**−0.028** **	**0.005**
	(0.139)	(0.012)	(0.009)	(0.150)	(0.012)	(0.009)	(0.145)	(0.012)	(0.009)

	Kernel Epan Bandwith(0.06)			Nearest Neighbor 5			Radius Caliper (0.01)		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	CESD	Depression	Happiness	CESD	Depression	Happiness	CESD	Depression	Happiness
control variables	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
i.island	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
constant	9.236***	0.330***	0.546***	9.281***	0.320***	0.532***	9.278***	0.336***	0.546***
	(0.586)	(0.047)	(0.041)	(0.620)	(0.050)	(0.043)	(0.597)	(0.048)	(0.042)
Observations	29,400	29,400	29,400	28,900	28,900	28,900	29,392	29,392	29,392
R^2	0.100	0.071	0.046	0.102	0.070	0.046	0.100	0.070	0.046

The bolded coefficient represents the DID estimator (betadid)

Standard errors in parentheses

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

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